

Biology, 2018 EC, entrance exam, version 5

Grade 12 · Biology · 2018 E.C.

Time allowed: 120 minutes

Questions: 97

Total marks: 97

Name _____ Grade _____ ID No. _____ Date _____

1. Professor Tilahun Yilma is well known internationally for his work on (1)
 - A. developing molluscicide
 - B. to safeguard biodiversity
 - C. developing a vaccine for rinderpest
 - D. developing commercial hybrid strain of sorghum
2. Which of the following research institute carries out research on leprosy, tuberculosis, leishmaniasis, malaria and HIV/AIDS? (1)
 - A. Armauer Hansen Research Institute (AHRI)
 - B. Institute of Biodiversity Conservation (IBC)
 - C. Akllu Lemma Institute of Pathobiology (ALIPB)
 - D. Institute of Agricultural Research (IAR)
3. Which one of the following is the disadvantage of electron microscope? (1)
 - A. It does not allow us to observe live specimen
 - B. It shows the inside details of a specimen
 - C. It shows us the three-dimensional shape of a specimen
 - D. It produces the image of the specimen on a screen
4. Electron microscope differ from light microscope in that: (1)
 - A. Electron microscope has higher magnification and resolution power
 - B. Electron microscope uses a beam of light
 - C. Electron microscope enables us to observe specimens in a greater detail
 - D. All are correct answers
5. What is the function of ribosomes? (1)

A. Produces antibodies	B. Protein synthesis
C. Covers the cell	D. Carry out respiration
6. An animal cell is placed in a certain solution then the cell swells and gradually bursts, the solution is said to be (1)

A. Neutral	B. hypertonic
C. Hypotonic	D. Isotonic
7. Turgor pressure and plasmolysis happen in plant cells when (1)
 - A. the cell is hypotonic and the environment is hypertonic, respectively.
 - B. excess water enters into the cells, and much water is lost from the cells, respectively.
 - C. the cell and the environment are isotonic, and the environment is hypotonic, respectively
 - D. excess water is lost from the cells and excess water enters into the cells, respectively.

-
- 8.** Which of the following double sugar is found in milk? (1)
- | | |
|-------------------|----------------------|
| A. Lactose | B. Maltose |
| C. Sucrose | D. Cellobiose |
-
- 9.** Why do we need a balanced diet? To (1)
- | | |
|---|-------------------------------------|
| A. prevent infection by syphilis | B. gain excessive weight |
| C. become very tall | D. maintain a healthy growth |
-
- 10.** Microscopic, finger-like projections that are found on the inner walls of the small intestine are responsible for (1)
- | | |
|---------------------|----------------------|
| A. ingestion | B. excretion |
| C. digestion | D. absorption |
-
- 11.** Which of the following structures controls the outflow of gastric contents into the duodenum? (1)
- | | |
|-----------------------------|--------------------------------|
| A. Cardiac sphincter | B. Esophageal sphincter |
| C. Pyloric sphincter | D. Urethral sphincter |
-
- 12.** The process of converting lipid into fat droplets is called (1)
- | | |
|--------------------------|------------------------|
| A. mastication | B. peristalsis |
| C. emulsification | D. assimilation |
-
- 13.** In which structure of the lungs does exchange of gases with the blood vessels takes place? (1)
- | | |
|--------------------|-----------------------|
| A. Alveoli | B. Trachea |
| C. Bronchus | D. Bronchioles |
-
- 14.** Which of the following is correct about the effect of altitude on breathing rate? At higher altitude breathing rate (1)
- A.** increases because of the low level of oxygen in the air.
- B.** decreases because of the atmospheric pressure.
- C.** increases because of the high level of carbon dioxide in the air.
- D.** decreases because of the high level of oxygen in the air.
-
- 15.** The volume of air inhaled and exhaled is called (1)
- | | |
|--------------------------|-----------------------|
| A. vital capacity | B. ventilation |
| C. tidal volume | D. oxygen debt |
-
- 16.** Which one of the following statements is correct about aerobic respiration? It (1)
- A.** produces lactic acid or alcohol as a byproduct.
- B.** produces less ATP than anaerobic respiration.
- C.** breaks down glucose into alcohol and carbon dioxide.
- D.** produces more ATP than anaerobic respiration.
-
- 17.** Identify the correct statement about ATP. (1)
- A.** ATP is used during diffusion.
- B.** ATP is the byproduct of cellular respiration.
- C.** Osmosis requires the use of ATP.
- D.** ATP is the energy currency of the cell.

-
- 29.** Seeds that are covered with fleshy/juicy fruits are most likely to be dispersed by (1)
- | | |
|------------|----------|
| A. animals | B. water |
| C. wind | D. fire |
-
- 30.** The termination of the dormancy period of a seed and onset of active growth of seed-embryo is termed as (1)
- | | |
|------------------|----------------|
| A. germination | B. pollination |
| C. fertilization | D. maturation |
-
- 31.** Stacks of thylakoid membrane are called (1)
- | | |
|------------|-----------|
| A. stroma | B. lumen |
| C. lamella | D. granum |
-
- 32.** Which of the following reactions does NOT occur during light-dependent reaction? (1)
- A. Carbon fixation or combining carbon dioxide with water.
 - B. Water molecule splits into hydrogen and oxygen.
 - C. Formation of ATP
 - D. Reduction of NADP+
-
- 33.** If you put iodine solution on a certain type of food and a blue-black color appeared. This is an indication the food consists of (1)
- | | |
|------------|-----------|
| A. protein | B. starch |
| C. mineral | D. lipid |
-
- 34.** Which of the following mechanisms enable the plant to transport water from the soil to the root and all the way up to the top of the plant? (1)
- | | |
|-------------------------|-------------------------|
| A. Translocation stream | B. Transduction stream |
| C. Transfusion stream | D. Transpiration stream |
-
- 35.** The root is (1)
- | | |
|---------------------------|---------------------------|
| A. positively hydrotropic | B. positively phototropic |
| C. negatively geotropic | D. negatively hydrotropic |
-
- 36.** Identify the correct statement about the properties of water. (1)
- A. Water is a non-polar molecule.
 - B. Water is an effective solvent for non-polar molecules.
 - C. Water has high specific heat capacity.
 - D. Water has the lowest density at 4°C
-
- 37.** Which of the following inorganic ions is the essential component of hemoglobin and myoglobin? (1)
- | | |
|------------|-----------|
| A. Calcium | B. Iron |
| C. Iodine | D. Sodium |
-
- 38.** In animals, extra glucose is stored in the form of (1)
- | | |
|-------------|-------------|
| A. starch | B. glucagon |
| C. glycogen | D. collagen |

39. Which of the following sequences represent the correct change in number of chromosomes during fertilization? (1)

- A. $2n \rightarrow n$
B. $2n \rightarrow n + n$
C. $n \rightarrow 2n$
D. $n + n \rightarrow 2n$

40. Identify the correct combination of technology and its imitation? (1)

- A.** Photo camera-human eye **B.** Bullet train-cucumber fruit
- C.** Robotic arm- claws of animals **D.** Airplane- reef shark

41. Which of the following statement best describe the benefits of biology to technology? (1)

- A. Many biochemical substances in plants used to develop medicine.
- B. Computer tomography is a device used to take many X-ray pictures.
- C. GPS is a device used to collect information on landscape mapping.
- D. A and B are correct answers

42. Which one of the following is the negative impact of technology on society and natural environment? (1)

- A. Improving the health conditions of humans.
B. Ensuring food security of humans.
C. Pesticides and fertilizers cause toxicity.
D. Providing energy and clean water.

43. Animals having variable body temperature are called (1)

- A. poikilotherms
B. warm-blooded
C. homeotherms
D. endothermic

44. Which of the following statements best describe characteristics of invertebrate animals? (1)

- A. They can maintain a constant body temperature.
- B. They possess internal rigid skeletal system.
- C. They constitute small portions of all animal species known.
- D. They are the most diverse group of animals known to humans.

45. Identify the correct statement that describes the characteristics of vertebrate animals. (1)

- A.** Vertebrates have centralized nervous system.
- B.** Vertebrates have endoskeleton made of bones.
- C.** Vertebrates have closed circulatory system
- D.** All are correct

46. Sexual reproduction differs from asexual reproduction in that: (1)

- A. offspring are exact copies of their parents
- B. brings genetic variation
- C. involves only one parent
- D. gamete formation is not required

47. Asexual reproduction is common among (1)

- A.** vertebrate animals **B.** endothermic animals
C. invertebrate animals **D.** warm-blooded animals

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- 48.** Parthenogenesis is a process by which (1)
- A. individuals develop from unfertilized egg
 - B. male gamete fuses with female gamete
 - C. only mitosis is involved in forming the male gamete
 - D. B and C are correct answers
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- 49.** Which of the following organisms carry out external fertilization? (1)
- A. Birds and Mammals
 - B. Insects and Reptiles
 - C. Honey bee and House fly
 - D. Frogs and Fishes
-
- 50.** Identify the correct statement about complete metamorphosis. (1)
- A. After the eggs hatched the nymph undergo a series of molting process.
 - B. The larval stage is its feeding stage.
 - C. It consists of three stages
 - D. Grasshopper and Mayflies undergo complete metamorphosis
-
- 51.** Which of the following statement is INCORRECT about reproduction of frogs? (1)
- A. Frogs larval stage is called a tadpole.
 - B. Tadpoles breathe using their gills.
 - C. Tadpoles are exclusively aquatic.
 - D. Tadpoles are similar to the adult frog.
-
- 52.** Which of the following sequences show the correct order of reproduction process in birds? (1)
- A. Mating → Egg laying → Incubation → Hatching
 - B. Incubation → Mating → Egg laying → Hatching
 - C. Egg laying → Mating → Hatching → Incubation
 - D. Egg laying → Hatching → Mating → Incubation
-
- 53.** Which of the following statement is NOT correct about rats? (1)
- A. Rats are mammals
 - B. They exhibit internal fertilization
 - C. They have one ovary
 - D. Fertilization takes place in the oviduct
-
- 54.** Identify the correct statement that describes the beneficial aspects of insects. (1)
- A. Some insects destroy crops and lead to food shortage.
 - B. Some insects serve as vectors.
 - C. Some insects serve as pollinators.
 - D. Some insects are pests of plants, fruits, and grains in store.
-
- 55.** Wildebeest populations perform an annual migration. This behavior of wildebeest can be taken as a good example of: (1)
- A. behavioral cycle
 - B. reproductive behavior
 - C. social behavior
 - D. competition cycle
-
- 56.** Which of the following statement is correct regarding blood-glucose level regulation in humans? (1)
- A. Low blood glucose level initiates secretion of glucagon.
 - B. Secretion of insulin raises blood glucose level.
 - C. Insulin secretion stimulates glycogen breakdown into glucose.
 - D. Glucagon stimulates glucose uptake from the blood.

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57. Which one of the following mechanisms can be considered as physiological means of temperature regulation? (1)
- A. Sweating
B. Hibernation
C. Clothing
D. Moving into sheds
-
58. Which of the following is the function of helicase enzymes? (1)
- A. It is responsible for forming new copies of DNA.
B. It breaks down neurotransmitters.
C. It is used to breakdown hexose sugars.
D. It separates the intertwined DNA strands.
-
59. The sequences of amino acids in the polypeptide chain is (1)
- A. secondary structure
B. primary structure
C. quaternary structure
D. tertiary structure
-
60. The primary difference between competitive and non-competitive inhibition is that: (1)
- A. competitive inhibition affects the substrate's ability to bind by binding an inhibitor in place of inhibitor.
B. competitive inhibitors bind to allosteric site of the enzyme.
C. in competitive inhibition, the inhibitors bind to the active site of the enzyme and prevent enzyme-substrate complex formation.
D. competitive inhibition allows the substrate bind to enzyme-inhibitor complex.
-
61. A class of enzyme called lyases are used for: (1)
- A. joining of two molecules.
B. splitting bonds using water.
C. breaking of various bonds by means other than hydrolysis.
D. transfer of functional group from one donor to acceptor molecule.
-
62. Which of the following statements correctly describes factors affecting enzymatic action? (1)
- A. Decreasing temperature below the optimum level results in denaturation of the enzyme molecule.
B. Temperature, inhibitors, end-product and substrate concentration are some of the factors affecting enzyme action.
C. Significant change in pH does not have an effect on the performance of enzymes.
D. Increasing substrate concentration beyond the optimum level increases the rate of reaction resulting in higher enzyme action.
-
63. In enzyme kinetics, how does increasing substrate concentration generally affect the rate of product formation and enzyme activity, assuming enzyme concentration is constant? (1)
- A. The rate increases with substrate concentration until it reaches a maximum velocity, after which it levels off.
B. The reaction rate decreases as substrate concentration increases due to enzyme inhibition.
C. The rate remains constant regardless of substrate concentration.
D. The reaction rate decreases initially and then increases as the substrate concentration increases.
-
64. Select the wrong combination of enzyme with its function in industries? (1)
- A. Papain- tendering meat
B. Rennin-cheese production
C. Ligninase-food production
D. Mannanase-removes stain from clothes

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- 65.** Which of the following statements is correct about malting process? (1)
- A. Germinating stage refers to planting the grain in the soil.
 - B. Killing stage involves roasting the beans or grains until it turns brown or black.
 - C. Steeping is a stage where soaking the grains or beans in water.
 - D. Germination is the last stage in the process of malting.
-
- 66.** Identify the INCORRECT statement about genetics. (1)
- A. Chromosomes that exist in pairs are called analogous pairs.
 - B. Gene is a small section of DNA that controls a specific trait.
 - C. Chromosomes are thread like structure made of DNA.
 - D. Some viruses use RNA as their genetic material.
-
- 67.** The process of exchanging segments of genetic material between two chromosomes or bivalents during prophase I of meiosis is called (1)
- A. cytokinesis
 - B. crossing over
 - C. karyokinesis
 - D. duplication
-
- 68.** Which of the following statements correctly describes what happens during translation stage? (1)
- A. Unwinding of DNA strands.
 - B. Conversion of the codes on mRNA into proteins.
 - C. Formation of mRNA from DNA.
 - D. Breakdown of proteins into amino acids.
-
- 69.** Conservation of natural resources includes all, EXCEPT (1)
- A. Preserving resources for future use.
 - B. Wise use of natural resources
 - C. Extensive use of natural resources.
 - D. Protection of natural resources.
-
- 70.** What are the possible consequences of deforestation of the tropical rain forest? (1)
- A. Reduction in species diversity of an area.
 - B. An increase in existing ecological niche.
 - C. Increased removal of carbon dioxide from the atmosphere.
 - D. An increase in the amount of nitrogen in the soil.
-
- 71.** A condition in which all people have access to sufficient, safe, and nutritious food is termed as (1)
- A. Food security
 - B. Over nutrition
 - C. Nutrition security
 - D. Malnutrition
-
- 72.** Identify the correct statement. (1)
- A. Biology empowers conscious citizens
 - B. Conscious citizen biologists develop innovative and cost effective bio-based technologies
 - C. Biology has a vital role in creating conscious citizen
 - D. All can be answer
-
- 73.** The type of microorganism used in the fermentation of alcoholic drinks is (1)
- A. saccharomyces
 - B. fusarium
 - C. archaea
 - D. candida

74. Application of biology in biotechnology can be explained in the making of (1)

- A.** Dairy products **B.** Antibiotics
C. Biosensors **D.** All can be answered

75. Which of the following statements is correctly describes bacterial cell? (1)

- A. Multicellular and Eukaryotic**
B. Single celled and Prokaryotic
C. Single celled and Eukaryotic
D. Multicellular and Prokaryotic

76. Adapting to extreme environment is the characteristics of (1)

- A. bacteria
B. virus
C. protozoa
D. archaea

77. Being eukaryotic, mostly filamentous, and chemoheterotrophic in terms of nutrition is likely the characteristics of (1)

- A. archaea
B. fungi
C. bacteria
D. virus

78. Which of the following disease is caused by protozoa? (1)

- A. Rabies
B. Giardiasis
C. Influenza
D. Chicken pox

79. Identify the INCORRECT combination of viral structure and its function. (1)

- A. Capsid** - protein coat
- B. Envelope** - surrounding nucleocapsid
- C. Plasmid** - small circular DNA
- D. Core** - genetic material

80. Which of the following statements is NOT true about normal microbiota? (1)

- A.** Colonize the host and cause disease **B.** Also known as 'normal flora'
- C.** Found in different parts of the body **D.** Found in healthy individuals

81. Diseases like salmonellosis and botulism are transmitted from infected person to a healthy one through (1)

- A.** eating contaminated food **B.** direct contact
C. droplet infection **D.** sexual intercourse

82. Bacteria like *Lactobacillus*, *Lactococcus* and *Streptococcus* play an important role in (1)

- A.** the production of insulin **B.** the recycling of minerals
- C.** food production and processing **D.** methane production and processing

83. Which of the following is the most effective method of killing pathogenic microorganisms? (1)

- A. Pasteurization
B. Boiling
C. Sanitization
D. Sterilization

84. A fungal infection that affects child's scalp and hair is (1)

- A.** Tinea corporis **B.** Tinea unguim
C. Tinea pedis **D.** Tinea capitis

85. The process of breaking down large and complex molecules through enzyme mediated reaction is termed as (1)

- A. catabolism
B. anabolism
C. physical digestion
D. churning

-
- 86.** Which one of the following is the main source from which plants get nutrients necessary for their growth and development? (1)
- A. Chlorophyll
B. Light
C. Atmosphere
D. Soil
-
- 87.** The interior of the chloroplast, called stroma is a site where: (1)
- A. photolysis of water takes place
B. light-independent stage of photosynthesis takes place
C. light-dependent stage of photosynthesis takes place
D. NADP⁺ is reduced to NADPH
-
- 88.** What is the advantage of a photosystem containing molecules of different types of light-sensitive pigments? It is to: (1)
- A. increase the size of the photosystem.
B. increase the surface area for light absorption
C. absorb light of different wavelengths
D. increase complexity of the photosystem
-
- 89.** What happens during light-dependent reaction of photosynthesis? (1)
- A. ATP is produced and NADP⁺ is reduced.
B. Carbon fixation
C. Glucose synthesis
D. NADPH is oxidized
-
- 90.** Photorespiration involves the oxidation of: (1)
- A. glucose
B. NADPH
C. CAM
D. RuBP
-
- 91.** Which one of the following statements correctly describes the contribution of photosynthesis for the continuity of life? (1)
- A. Photosynthesis releases additional carbon to the atmosphere.
B. Photosynthesis provides oxygen and food for heterotrophs.
C. Photosynthesis contributes greenhouse gases to the atmosphere.
D. Photosynthesis absorbs oxygen from the atmosphere.
-
- 92.** Cellular respiration is the process by which cells: (1)
- A. breakdown glucose to produce ATP
B. utilize ATP to produce glucose
C. breakdown ADP to form AMP and P
D. all are correct answers
-
- 93.** Which of the following steps in cellular respiration does NOT require the use of oxygen? (1)
- A. Krebs cycle
B. Glycolysis
C. Chemiosmosis
D. Link reaction
-
- 94.** Much of the ATP is produced during: (1)
- A. Glycolysis
B. oxidative phosphorylation
C. link reaction
D. Krebs cycle
-
- 95.** Which of the following is NOT the product of Krebs cycle? (1)
- A. Pyruvate
B. CO₂
C. NADH
D. ATP

96. Substrate level phosphorylation takes place in:

(1)

- | | |
|--|---|
| A. Link reaction and electron transport | B. Glycolysis and Krebs cycle |
| C. Link reaction and Krebs cycle | D. Glycolysis and electron transport chain |
-

97. During alcohol fermentation, pyruvic acid in the presence of pyruvate decarboxylase is first converted to

(1)

- | | |
|------------------------|-------------------|
| A. Acetaldehyde | B. Lactose |
| C. Lactic acid | D. Glucose |

— END OF EXAMINATION —

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